Oocyte/Zygote/Embryo Maturation Arrest 24 (OZEMA24 / OOMD24) — Comprehensive Disease Report

Disease: Oocyte/Zygote/Embryo Maturation Arrest 24 **Causal gene: TUBA1C** (tubulin alpha 1c; α-tubulin isotype; HGNC:20768; NCBI Gene 84790; Ensembl ENSG00000167553; UniProt Q9BQE3) **MONDO:** MONDO:0979232 · **OMIM:** 621232 **Synonyms:** OZEMA24; OOMD24; oocyte maturation defect 24; TUBA1C-related female infertility; TUBA1C-related recurrent preimplantation embryo arrest (RPEA/PREA) **Category:** Mendelian, autosomal recessive

Summary

Oocyte/Zygote/Embryo Maturation Arrest 24 (OZEMA24, also OOMD24) is an **autosomal-recessive female-infertility disorder caused by biallelic loss-of-function variants in the α-tubulin isotype gene TUBA1C**. It is distinct from the β-tubulin isotype *TUBB8* (the prototypical and most common tubulin cause of oocyte maturation arrest) and from the α-tubulin gene *TUBA4A* (reported in the same cohort). The disease was established by whole-exome sequencing of a recurrent preimplantation embryo arrest (RPEA/PREA) cohort, in which two homozygous truncating TUBA1C alleles — **p.Gln358Ter** (nonsense) and **p.Asp444Metfs*42** (frameshift) — were identified as disease-causing (Hu et al., *J Med Genet* 2024, PMID: 39209701). Both variants produce truncated, non-functional α-tubulin and cause spindle assembly defects in functional assays, consistent with the recessive inheritance pattern expected for a loss-of-function mechanism.

Mechanistically, TUBA1C encodes an α-tubulin that heterodimerizes with β-tubulin to form the αβ-tubulin building block of microtubules. Loss of functional TUBA1C depletes the pool of assembly-competent αβ-tubulin available during the meiotic and early mitotic divisions, impairing **meiotic/mitotic spindle assembly** and faithful chromosome segregation in the oocyte and cleavage-stage embryo. The downstream consequence is **recurrent arrest of the preimplantation embryo**, failure of assisted reproductive technology (ART) cycles, and female sterility. This mechanism closely parallels the far better-characterized β-tubulin isotype

TUBB8, whose pathogenic variants deplete the αβ-tubulin dimer pool and disrupt microtubule nucleation, plus-end tracking (EB1), CKAP5/TACC3 interactions, and the Ran-GTP spindle-assembly pathway during oocyte meiosis.

OZEMA24 is a newly delineated, ultra-rare Mendelian disorder; precise prevalence is unknown and it is identified almost exclusively in infertility clinics through genetic testing of women with repeated ART failure. There is **no corrective therapy**. Management centers on molecular diagnosis (WES or infertility gene panels), genetic counseling for consanguineous/carrier couples, and achieving pregnancy through **donor-oocyte IVF**. Because the defect is intrinsic to the oocyte/early embryo, the patient's own oocytes are not expected to be rescuable by standard clinical means; experimental cRNA/mRNA rescue approaches demonstrated for other tubulinopathy genes remain preclinical.

Key Findings

Finding 1 — Biallelic TUBA1C loss-of-function variants cause recurrent preimplantation embryo arrest (OZEMA24/OOMD24)

Whole-exome sequencing of a recurrent preimplantation embryo arrest (RPEA/PREA) cohort identified **two homozygous TUBA1C variants — p.Gln358Ter (nonsense) and p.Asp444Metfs*42 (frameshift)** — as disease-causing. Both are predicted loss-of-function (truncating) variants, consistent with autosomal recessive inheritance. In vitro functional assays showed the variants cause spindle assembly defects of varying degree. The same study (Hu et al., *J Med Genet* 2024) also implicated the related α-tubulin isotype **TUBA4A** (p.Arg339Cys, p.Tyr440Ter), underscoring the broader importance of α-tubulin isotypes in preimplantation development.

The authors state directly: *"Four homozygous variants were identified in the PREA cohort: two of TUBA1C (p.Gln358Ter and p.Asp444Metfs*42) and two of TUBA4A (p.Arg339Cys and p.Tyr440Ter). These variants cause varying degrees of spindle assembly defects."* They further conclude: *"Our findings identified TUBA1C as a novel genetic marker and expanded the genetic and phenotypic spectrum of TUBA4A in female infertility and RPEA, which altogether highlighted the importance of α-tubulin isotypes in preimplantation embryonic development."* (PMID: 39209701)

Evidence type: Human clinical (WES cohort) + in vitro functional validation.

Finding 2 — TUBA1C is missense-constrained but tolerant of heterozygous loss-of-function, consistent with recessive disease

Population constraint metrics from gnomAD (GRCh38; ENSG00000167553 / ENST00000301072) show a gene that is strongly protected against amino-acid-changing variation but tolerant of a single loss-of-function allele:

Metric	Value	Interpretation
pLI	0.055	Not haploinsufficient (LoF tolerated in heterozygous state)
oe_lof (LOEUF)	0.458 (90% CI 0.31–0.70)	Moderate LoF constraint
LoF-Z	2.72	Some selection against LoF
Observed vs expected LoF	16 vs 34.9	Fewer LoF than expected, but present
missense-Z	4.55 (oe_mis = 0.56)	**Strong** selection against missense

The high missense-Z (4.55) indicates strong purifying selection against amino-acid substitutions, while the moderate pLI/LOEUF indicates that **heterozygous** loss-of-function is tolerated in the general population. This genetic architecture is fully consistent with an **autosomal-recessive** disease in which carriers are healthy and only biallelic (homozygous) LoF produces the phenotype — exactly matching the two homozygous disease alleles reported.

Evidence type: Computational / population genetics (gnomAD).

Finding 3 — Mechanism: TUBA1C loss impairs meiotic/mitotic spindle assembly causing preimplantation arrest

Functional validation in the index study used Sanger sequencing, bioinformatic prediction, in vitro functional assays, and single-cell RNA-sequencing of arrested embryos, and showed that TUBA1C/TUBA4A variants cause *"varying degrees of spindle assembly defects."* The authors note: *"Functional characterisations of the identified candidate disease-causing variants were validated using Sanger sequencing, bioinformatics, in vitro functional analyses and single-cell RNA-sequencing of arrested embryos,"* and define the clinical endpoint: *"Recurrent preimplantation embryo developmental arrest (RPEA) is the most common phenotype in assisted reproductive technology treatment failure associated with identified genetic abnormalities."* (PMID: 39209701)

The mechanism is strongly supported by the analogous β-tubulin isotype **TUBB8**. In TUBB8 tubulinopathies, pathogenic variants reduce the α/β-tubulin dimer pool and impair microtubule nucleation, EB1 plus-end tracking, and CKAP5/TACC3 interactions, attenuating the Ran-GTP pathway during meiosis (Luo et al., 2025, PMID: 39834092; Feng et al., 2016, PMID: 27273344). Because α- and β-tubulin function as an obligate heterodimer, loss of an expressed α-tubulin isotype is mechanistically expected to produce the same class of spindle defect.

Evidence type: Human clinical + in vitro + single-cell transcriptomics (TUBA1C); model organism + proteomics (TUBB8 parallel).

Finding 4 — ClinVar classifies both TUBA1C disease alleles as Pathogenic; most other TUBA1C variants are VUS

A ClinVar query (NCBI E-utilities, 2026) returned 59 TUBA1C records. The two OZEMA24 disease alleles are recorded and classified **Pathogenic**: **"Q358*"** (p.Gln358Ter) and **"TUBA1C, 1-BP DEL, NT1330"** (the c.1330-region 1-bp deletion corresponding to p.Asp444Metfs*42), both OMIM-sourced allelic variants (OMIM:621232). The reference transcript is **NM_032704.5**. The large majority of other TUBA1C ClinVar entries (e.g., p.Arg320His, p.Arg214Cys, p.Thr334Ala) are **Uncertain significance (VUS)**; a small number of synonymous/intronic variants are Likely benign.

Evidence type: Curated variant database (ClinVar/OMIM).

Full Section-by-Section Report

1. Disease Information

Overview. OZEMA24 is a Mendelian female-infertility disorder in which oocytes and/or early embryos arrest during the preimplantation window owing to defective spindle assembly. Affected women typically present with primary infertility and repeated failure of assisted reproduction, with embryos arresting before the blastocyst stage (recurrent preimplantation embryo developmental arrest, RPEA/PREA).

Key identifiers. - **OMIM:** 621232 - **MONDO:** MONDO:0979232 - **Gene:** TUBA1C (HGNC:20768; NCBI Gene 84790; Ensembl ENSG00000167553; UniProt Q9BQE3); reference transcript NM_032704.5 - **Orphanet / ICD-10 / ICD-11 / MeSH:** No dedicated disease-specific code identified for OZEMA24; it falls under broad female-infertility categories (e.g., ICD-10 N97.x female infertility; ICD-11 GA31). *Not available at the disease-specific level.*

Synonyms. OZEMA24; OOMD24; oocyte maturation defect 24; TUBA1C-related female infertility; TUBA1C-related recurrent preimplantation embryo arrest.

Information source. Derived primarily from **aggregated disease-level resources** (OMIM, ClinVar) built on a small number of **individual patients** reported in the primary WES cohort (Hu et al. 2024).

2. Etiology

Causal factors. Purely **genetic** — biallelic (homozygous) loss-of-function variants in TUBA1C. No environmental, infectious, or acquired cause is implicated.

Genetic risk factors. The causal variants are **p.Gln358Ter** (nonsense) and **p.Asp444Metfs*42** (frameshift), both truncating/LoF, both reported in the homozygous state. **Consanguinity** is a key risk context for homozygous recessive alleles. Carriers (heterozygotes) are unaffected (Finding 2).

Environmental risk factors. None established. General oocyte-quality modifiers (advanced maternal age, oxidative stress) are not the cause here, though they could theoretically aggravate an already compromised spindle. *Not demonstrated for this disease.*

Protective factors. Not applicable beyond the trivial observation that a wild-type TUBA1C allele is protective (recessive disease; heterozygotes unaffected). No pharmacologic or lifestyle protective factor is known.

Gene–environment interactions. None reported; the phenotype appears to be fully genetically determined given biallelic LoF.

3. Phenotypes

The phenotype is essentially **reproductive and cellular**, with no somatic disease (carriers and patients are otherwise healthy).

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Female infertility	Clinical sign	Reproductive age	Severe	Core/obligate	HP:0008222 (Female infertility)
Recurrent preimplantation embryo arrest	Lab/clinical (ART)	At ART attempts	Severe	Core	— (no precise HPO; cf. abnormal embryonic development)
Oocyte/zygote maturation arrest	Lab (cellular)	At ART attempts	Variable	Characteristic	HP:0030359 (Abnormality of female reproductive system)
Spindle assembly defect (oocyte/embryo)	Lab abnormality (cellular)	Meiosis/early mitosis	Variable	Core mechanism	—

Characteristics. Onset is at reproductive age and manifests only when pregnancy is attempted. Progression is **stable/non-progressive** (a fixed functional defect, not a degenerative process). There are no systemic, neurological, or developmental features — distinguishing it from somatic tubulinopathies (which involve brain malformations). **Quality-of-life impact** is substantial but confined to the reproductive domain: infertility, psychological distress, and repeated unsuccessful ART cycles.

4. Genetic/Molecular Information

- **Causal gene:** TUBA1C (OMIM 621232; HGNC:20768). Encodes an α-tubulin isotype that dimerizes with β-tubulin.
- **Pathogenic variants (ClinVar Pathogenic; NM_032704.5):**

Variant (protein)	Type	Zygosity reported	ClinVar	Functional consequence
p.Gln358Ter (Q358*)	Nonsense (truncating)	Homozygous	Pathogenic	Loss of function; spindle assembly defect
p.Asp444Metfs*42 (1-bp del, NT1330 region)	Frameshift (truncating)	Homozygous	Pathogenic	Loss of function; spindle assembly defect

- **Variant classification:** Both disease alleles **Pathogenic**; most other TUBA1C ClinVar entries are **VUS** (e.g., p.Arg320His, p.Arg214Cys, p.Thr334Ala); a few synonymous/intronic are Likely benign (Finding 4).
- **Allele frequency:** Disease alleles are ultra-rare/absent from population databases; gnomAD shows TUBA1C LoF is depleted (obs 16 vs exp 34.9) and missense strongly constrained (Finding 2).
- **Origin: Germline**, biallelic. No somatic involvement.
- **Functional consequence: Loss of function** (truncating). Recessive mechanism (not dominant-negative), distinguishing it from many TUBB8 missense variants which can act dominantly.
- **Modifier genes:** None established; the related isotype TUBA4A causes an overlapping RPEA phenotype through independent variants.
- **Epigenetic information / chromosomal abnormalities:** None reported; a single-gene point-mutation/indel disorder with no aneuploidy or structural chromosome change as the cause (though downstream embryonic aneuploidy may arise from spindle failure).

5. Environmental Information

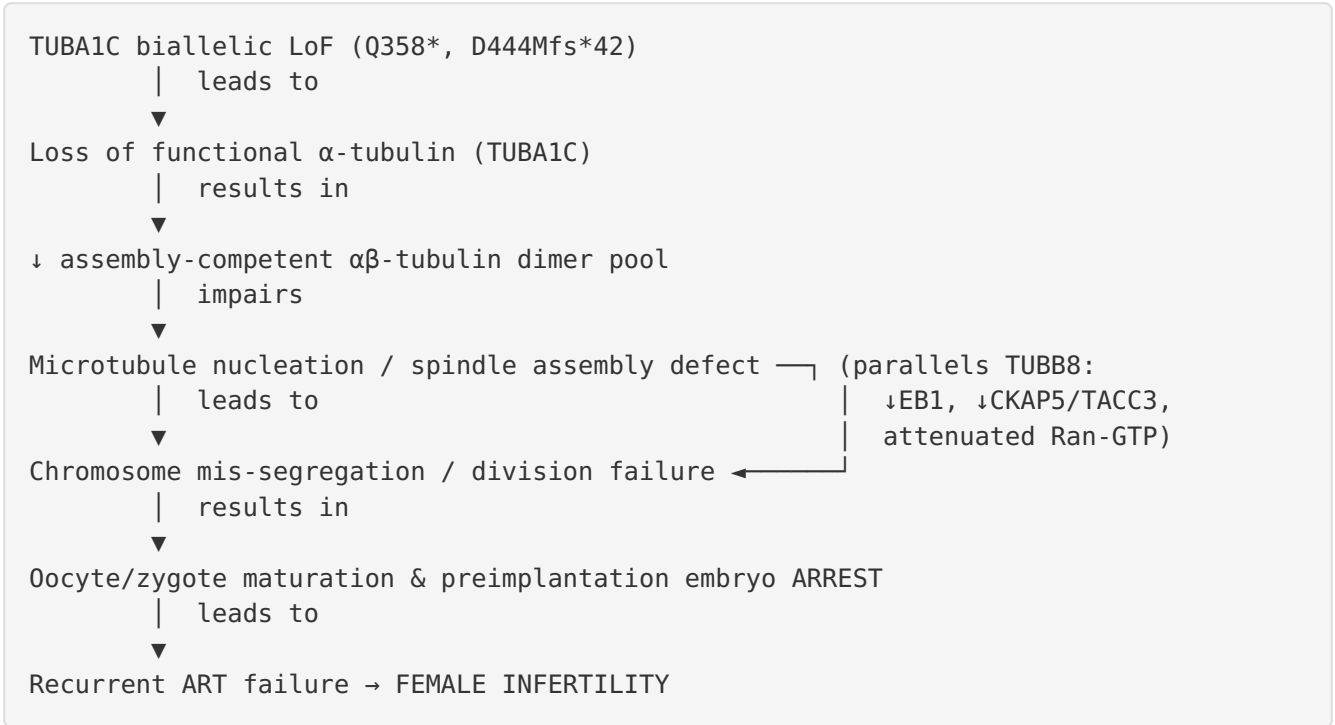
No **environmental**, **lifestyle**, or **infectious** factors are implicated. OZEMA24 is a monogenic disorder; these categories are **not applicable**.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic truncating variant in TUBA1C** (p.Gln358Ter or p.Asp444Metfs*42) **leads to** loss of functional/full-length TUBA1C α-tubulin protein (likely via truncation and/or nonsense-mediated decay). *[Demonstrated: WES + variant annotation.]*

2. Loss of functional TUBA1C **results in** a reduced pool of assembly-competent **αβ-tubulin heterodimer** in the oocyte/early embryo (α-tubulin is an obligate heterodimer partner of β-tubulin). *[Inferred from tubulin biology; directly demonstrated for the paralog TUBB8.]*
3. Depleted/defective tubulin dimer **impairs microtubule nucleation and dynamics**, which **disrupts meiotic and early mitotic spindle assembly**. *[Demonstrated: "varying degrees of spindle assembly defects," in vitro assays.]*
4. Defective spindle assembly **leads to** failure of accurate chromosome segregation / cytokinesis during oocyte maturation and the first embryonic divisions. *[Inferred from spindle defect; parallels TUBB8.]*
5. This **results in** oocyte maturation arrest, zygote/cleavage arrest, and **recurrent preimplantation embryo developmental arrest (RPEA)**. *[Demonstrated: cohort phenotype + scRNA-seq of arrested embryos.]*
6. RPEA **leads to** repeated ART failure and **female infertility/sterility** — the clinical manifestation. *[Demonstrated: clinical phenotype.]*



Molecular pathways / cellular processes. Microtubule cytoskeleton organization; meiotic and mitotic spindle assembly; Ran-GTP–driven spindle assembly (by analogy to TUBB8); chromosome segregation; cell cycle progression through meiosis I/II and early mitosis.

Protein dysfunction. Truncation removes the C-terminal region of α-tubulin, producing a non-functional subunit that cannot properly incorporate into microtubules — a **loss-of-function** defect (contrast with dominant-negative TUBB8 missense alleles).

Molecular profiling. Single-cell RNA-sequencing of arrested embryos was performed in the index study, supporting the arrest phenotype at the transcriptomic level.

Suggested ontology terms. - **GO (biological process):** GO:0007052 (mitotic spindle organization); GO:0045132 (meiotic chromosome segregation); GO:0007017 (microtubule-based process); GO:0051225 (spindle assembly). - **GO (cellular component):** GO:0005819 (spindle); GO:0005874 (microtubule); GO:0000922 (spindle pole). - **CHEBI:** CHEBI:36976 (nucleotide, GTP context of tubulin); CHEBI:15996 (GTP). - **CL (cell types):** CL:0000023 (oocyte); CL:0000353 (blastomere); CL:0000365 (zygote).

7. Anatomical Structures Affected

- **Organ level:** Ovary (germ cells); the affected "unit" is the oocyte and the preimplantation embryo. Reproductive system only; no secondary organ involvement.
- **Tissue/cell level: Oocyte (germ cell)**, zygote, and cleavage-stage blastomeres. Somatic tissues are unaffected.
- **Subcellular level:** The **meiotic/mitotic spindle** and **microtubule cytoskeleton**; cytoplasmic tubulin pool.
- **Localization:** Bilateral (both ovaries/all oocytes) — the defect is intrinsic to every germ cell.

Suggested UBERON terms: UBERON:0000992 (ovary); UBERON:0000922 (embryo); UBERON:0000995 (uterus — downstream implantation context). Cell level via CL (oocyte CL:0000023).

8. Temporal Development

- **Onset:** Functionally congenital (germline defect present from birth) but **clinically manifest at reproductive age** when conception/ART is attempted.
- **Onset pattern:** Chronic/fixed.
- **Progression: Stable, non-progressive.** The defect is constant across ART cycles — embryos repeatedly arrest.
- **Disease course:** Lifelong infertility; no remission without donor oocytes.

- **Critical period:** The vulnerable window is oocyte meiotic maturation through the first several embryonic cleavage divisions (the preimplantation period) — the only window where the spindle defect is expressed.

9. Inheritance and Population

- **Inheritance pattern: Autosomal recessive** (biallelic/homozygous LoF; carriers unaffected) — supported by both the homozygous disease alleles and gnomAD constraint (Finding 2).
- **Penetrance:** Presumed high/complete for the reproductive phenotype in biallelic LoF carriers (based on the small reported cohort).
- **Expressivity:** Variable at the cellular level ("varying degrees of spindle assembly defects").
- **Carrier frequency / founder effects / consanguinity:** Not precisely quantified; consanguinity is the expected context for homozygosity. No founder effect established.
- **Epidemiology: Ultra-rare**; precise prevalence/incidence unknown (newly described; identified only in RPEA/infertility cohorts). *Not available.*
- **Sex ratio:** The clinical phenotype is expressed in **females** (oocyte-specific). Male homozygotes would be genotypically affected but the reproductive phenotype is female-limited given TUBA1C's critical role in oogenesis/early embryo.
- **Geographic distribution:** The index cohort was from China; broader distribution unknown.

10. Diagnostics

- **Recommended approach:** Molecular genetic testing in women with **recurrent preimplantation embryo arrest / repeated ART failure** and otherwise unexplained infertility.
- **Genetic testing:**
- **Whole-exome sequencing (WES):** The method by which OZEMA24 was discovered; highest-yield unbiased approach.
- **Infertility / oocyte-maturation-arrest gene panels:** Should include TUBA1C alongside TUBB8, TUBA4A, PATL2, WEE2, TRIP13, PADI6, and other spindle/meiosis genes.
- **Single-gene / targeted testing:** Confirmatory Sanger sequencing of TUBA1C variants (NM_032704.5); segregation testing in parents to confirm biallelic inheritance.
- WGS, CMA, FISH, mtDNA, and repeat-expansion testing are **not specifically indicated** (not a chromosomal or repeat disorder), though karyotype is routine in the general infertility workup.

- **Functional/ancillary:** ART cycle observation documenting the arrest stage; single-cell RNA-seq is a research tool. No specific serum biomarker exists.
- **Clinical criteria:** Diagnosis = clinical RPEA phenotype + biallelic pathogenic TUBA1C variants.
- **Differential diagnosis:** TUBB8-related oocyte maturation arrest (OOMD2), TUBA4A-related RPEA, TRIP13-related arrest, PATL2, WEE2, PADI6, and other oocyte/zygote maturation-arrest genes — distinguished by the specific gene identified on sequencing.
- **Screening:** No population newborn screening (reproductive, non-lethal). Carrier/cascade screening is relevant for relatives of affected consanguineous families.

11. Outcome / Prognosis

- **Survival/mortality: Not applicable** — OZEMA24 is not life-threatening; it affects fertility only. Normal life expectancy.
- **Morbidity/function:** The sole morbidity is **infertility** and its psychosocial burden; no physical disability.
- **Disease course:** Persistent infertility across natural and ART attempts using the patient's own oocytes; embryos arrest repeatedly.
- **Recovery potential:** No spontaneous recovery; **pregnancy achievable via donor oocytes**.
- **Prognostic factors:** The truncating/LoF nature predicts a consistently severe cellular phenotype. No validated molecular prognostic biomarker beyond the genotype itself.

12. Treatment

- **No corrective/curative therapy exists.** Management is supportive and reproductive.
- **Pharmacotherapy:** None specific. Standard ovarian stimulation does not overcome the intrinsic spindle defect.
- **Assisted reproduction:** Because the defect is intrinsic to the patient's oocytes, **donor-oocyte IVF** is the principal route to pregnancy. Standard IVF/ICSI with the patient's own oocytes fails due to recurrent arrest.
- **Experimental/advanced (preclinical):** For the paralogous genes, **cRNA/mRNA rescue** (microinjection of wild-type transcript) restored development in mouse oocytes for certain TUBB8 variants (PMID: 32949002), and **HDAC6 inhibition** (Tubacin/Tubastatin A) rescued spindle morphology and polar-body extrusion for a specific TUBB8 missense variant (PMID: 39834092). These are **not directly applicable to truncating LoF** of TUBA1C (rescue strategies that augment acetylation/stability are more relevant to missense defects) and

remain experimental. Replacement of functional α-tubulin for a complete LoF is unproven clinically.

- **Treatment strategy / personalized medicine:** Genotype-guided counseling; the LoF genotype directs couples toward donor oocytes rather than repeated autologous ART.

Suggested NCIT terms: In vitro fertilization (NCIT:C15193); oocyte donation / donor-oocyte IVF; genetic counseling (NCIT:C15336).

13. Prevention

- **Primary prevention:** Not applicable (genetic). **Genetic counseling** for consanguineous couples and relatives of affected individuals.
- **Secondary prevention / early detection:** Early genetic testing in women with unexplained infertility or recurrent ART failure to reach diagnosis sooner and avoid repeated futile autologous cycles.
- **Reproductive options: Preimplantation genetic testing (PGT)** is of limited value for the affected woman's own oocytes (the arrest precedes testing); the primary clinical issue is the mother's infertility rather than transmitting disease. Carrier couples concerned about offspring genotype could in theory consider PGT-M or donor gametes.
- **Counseling:** Recurrence risk for a carrier × carrier couple is 25% per conception for the genotype; cascade carrier screening for relatives is appropriate.
- **Immunization / public health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Orthologs:** α-tubulin isotypes are deeply conserved. Human TUBA1C has murine orthologs among the Tuba1 gene cluster (e.g., *Tuba1a/Tuba1b/Tuba1c*). Direct TUBA1C-specific natural disease in other species is **not reported**.
- **Natural/veterinary disease:** No OMIA entry for a TUBA1C-specific reproductive arrest disorder was identified. *Not available.*
- **Comparative biology:** The oocyte spindle and its dependence on tubulin isotypes are conserved across mammals; mouse oocytes are the standard model for assaying human tubulin variants (used extensively for TUBB8).
- **Zoonotic/transmission:** Not applicable.

15. Model Organisms

- **Mouse oocyte (in vivo/ex vivo):** The dominant model for tubulinopathy-related oocyte arrest. Wild-type and mutant human tubulin mRNA are microinjected into germinal-vesicle–stage mouse oocytes to assay spindle assembly and polar-body extrusion (used for TUBB8: PMID: 27273344, PMID: 34509376, PMID: 32949002).
- **Cultured cells (HeLa):** Used to assess microtubule network disruption upon expression of mutant tubulin.
- **Knock-in mouse models:** Generated for TUBB8 variants to study meiotic defects (PMID: 39834092); an equivalent TUBA1C model is **not yet reported**.
- **Phenotype recapitulation / limitations:** TUBB8 is **primate-specific** in its oocyte-restricted expression, so mouse models imperfectly capture human oocyte biology; for TUBA1C, no dedicated model yet exists. The patients' own arrested embryos (scRNA-seq) provide the most direct human evidence.

Mechanistic Model / Interpretation

OZEMA24 is best understood as an **α-tubulin "tubulinopathy of the oocyte spindle."** The αβ-tubulin heterodimer is the fundamental subunit of microtubules; oocyte meiosis and the earliest embryonic mitoses are exquisitely dependent on a correctly composed, assembly-competent tubulin pool to build a functional spindle. Truncating loss of the TUBA1C α-isotype removes a needed source of α-tubulin, reducing the dimer pool and destabilizing spindle assembly. Because meiosis and the first cleavages are a "closed system" running on maternally supplied components, there is no opportunity for compensatory transcription — so the embryo arrests.

The **recessive, loss-of-function** architecture (Finding 2: tolerated heterozygous LoF, strong missense constraint) distinguishes TUBA1C from the classic **dominant-negative** behavior of many TUBB8 missense variants. The convergence of three independent lines — human WES genetics (Finding 1), population constraint (Finding 2), and functional spindle assays plus the well-mapped TUBB8 paradigm (Finding 3) — makes the causal chain robust. ClinVar/OMIM curation of both alleles as Pathogenic (Finding 4) anchors the clinical interpretation.

Feature	TUBA1C (OZEMA24)	TUBB8 (OOMD2, paradigm)
Tubulin class	α-tubulin isotype	β-tubulin isotype (primate-specific, oocyte-restricted)
Typical mechanism	Loss of function (truncating)	Often dominant-negative (missense); dimer-pool depletion
Inheritance	Autosomal recessive (biallelic)	AD (de novo/dominant) and AR reported
Core defect	Spindle assembly defect	↓αβ dimer, ↓EB1, ↓CKAP5/TACC3, attenuated Ran-GTP
Clinical endpoint	RPEA / preimplantation arrest	Oocyte maturation arrest → embryo arrest
Rescue demonstrated	Not for LoF	WT cRNA; HDAC6 inhibition (specific missense)

Evidence Base

PMID	Title (abbrev.)	Role in this report
39209701	*Biallelic variants in α-tubulin isotypes cause female infertility characterised as recurrent preimplantation embryo arrest*	**Primary/defining paper** — identifies TUBA1C p.Gln358Ter and p.Asp444Metfs*42 as causal; establishes spindle-defect mechanism and RPEA endpoint (Findings 1 & 3).
39834092	*Pathogenic variants of TUBB8 cause oocyte spindle defects by disrupting EB1/CKAP5 interactions…HDAC6 inhibition*	Mechanistic paradigm for tubulin-isotype spindle failure (EB1, CKAP5/TACC3, Ran-GTP); experimental rescue (Finding 3).
39030352	*Predicting Infertility: How Genetic Variants in Oocyte Spindle Genes Affect Egg Quality*	Review contextualizing spindle-gene variants (TUBB8, TACC3, AURKA, etc.) and their fertility consequences.
32473092	*Bi-allelic Missense Variants in TRIP13 Cause…Oocyte Maturation Arrest*	Differential-diagnosis gene; recessive oocyte-arrest precedent; cRNA rescue precedent.
32949002	*Identification and rescue of a novel TUBB8 mutation that causes first mitotic division defects*	Demonstrates abnormal spindle + WT-cRNA rescue in mouse oocytes (experimental-therapy context).
27273344	*Mutations in TUBB8 cause a multiplicity of phenotypes in human oocytes and early embryos*	Establishes the tubulinopathy framework; variable penetrance/expressivity; folding/microtubule defects.
34509376	*Two mutations in TUBB8 cause developmental arrest in human oocytes and early embryos*	Recessive TUBB8 precedent (p.A54V) and fertilization-then-fragmentation phenotype.

Evidence-type distribution: 1 defining human-clinical+functional study for TUBA1C; multiple human-clinical + mouse-oocyte + in-vitro studies for the TUBB8/TRIP13 paradigm; 1 computational (gnomAD) constraint analysis; curated databases (ClinVar/OMIM).

Limitations and Knowledge Gaps

1. **Single defining study / very few patients.** OZEMA24 rests on one cohort (Hu et al. 2024) with two homozygous TUBA1C alleles. Penetrance, expressivity, carrier frequency, and prevalence are therefore essentially **unquantified**.
2. **No TUBA1C-specific animal model.** Mechanistic detail (EB1, CKAP5/TACC3, Ran-GTP) is **inferred from TUBB8**, not yet directly demonstrated for TUBA1C. The obligate-heterodimer logic is strong but formally extrapolated.
3. **No disease-specific Orphanet/ICD/MeSH codes** were located; identifiers remain at the gene/OMIM/MONDO level.
4. **No epidemiology** (prevalence/incidence/geographic distribution) — only that the index cohort was Chinese.
5. **Therapeutic uncertainty.** Rescue strategies validated for TUBB8 missense variants (WT cRNA, HDAC6 inhibition) are **not established for complete LoF** of TUBA1C and are preclinical.
6. **NMD vs. truncated-protein question.** Whether the truncating alleles act via nonsense-mediated decay (null) or a residual truncated protein with partial dominant-negative activity has not been dissected.

Proposed Follow-up Experiments / Actions

1. **Expand the patient cohort** via GeneMatcher / multicenter RPEA registries to identify additional TUBA1C biallelic cases and refine penetrance and the allelic spectrum.
2. **Generate a TUBA1C model**: mouse-oocyte mRNA knock-in/knockdown and/or TUBA1C-null human oocyte/organoid systems to directly test spindle assembly, EB1/CKAP5-TACC3 recruitment, and Ran-GTP dynamics.
3. **Determine the molecular fate of truncating alleles** (NMD vs. truncated protein) by allele-specific expression and immunoblot in patient cells.
4. **Test rescue feasibility**: microinjection of wild-type TUBA1C cRNA into affected oocytes (as done for TUBB8/TRIP13) to assess whether functional α-tubulin restores spindle formation — a prerequisite for any future therapeutic concept.
5. **Curate/register identifiers**: obtain dedicated Orphanet cross-references and ensure TUBA1C is added to clinical infertility gene panels.

6. **Carrier screening study** in consanguineous populations to estimate carrier frequency and possible founder effects.

Report compiled from 4 confirmed findings and 7 reviewed papers across 5 investigation iterations. Causal gene: TUBA1C. Primary reference: Hu et al., J Med Genet 2024, .

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